NCAA March Madness Prediction System

Mixture-of-Experts, Temporal Modeling, and Prediction Market Trading

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Abstract

We propose a probabilistic prediction system for the NCAA Men's Basketball Tournament ("March Madness"), targeting both the Kaggle *March Machine Learning Mania* competition (2027 edition) and live trading on the Kalshi prediction market. Our approach combines three methodological innovations over standard baselines: (1) a **Mixture-of-Experts** architecture that routes matchups to specialized sub-models by game type, (2) a **Transformer sequence model** that encodes a team's full-season trajectory into a dense embedding, and (3) a **market-calibrated ensemble** that fuses model predictions with betting market prices via Bayesian updating. We backtest on 12 years of tournament data (2014–2026, ∼756 games) using leave-one-tournament-out cross-validation, targeting a Brier score below 0.170 (vs. ∼0.200 seed baseline and ∼0.175 Vegas benchmark).

1 Introduction

The NCAA Division I Men's Basketball Tournament is a 68-team, single-elimination event producing 63 games over three weeks each March. Its high variance—driven by single-elimination format, large talent gaps, and emotional intensity—makes it a canonical testbed for probabilistic prediction.

The Kaggle *March Machine Learning Mania* competition, running annually since 2014 with a $50,000 prize pool, asks participants to submit $P(\text{Team}_i \text{ beats Team}_j)$ for every possible matchup, scored by Brier score:

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}(p_i - o_i)^2 \tag{1}$$

where p_i is the predicted probability and $o_i \in \{0, 1\}$ is the outcome. Lower is better.

A key empirical finding from 12 years of competition: **simple models with strong features consistently outperform complex ML**. Most winners use logistic regression or Elo with KenPom efficiency ratings [Pomeroy, 2004]. The notable exception is Landgraf [2017], who won by modeling competitors' submissions (meta-prediction) rather than improving game-level prediction.

This project aims to push the prediction frontier through structured model heterogeneity (Mixture-of-Experts), temporal context (Transformer), and market integration (Kalshi), while respecting the fundamental constraint that 63 games per tournament provide very limited signal.

1.1 Connection to Prior Work

This project extends our prior NBA analytics research in `sports_project/`:

- **Context-Weighted RAPM** (CW-RAPM): 1.9% RMSE improvement over standard RAPM via stint-level quality weighting — analogous context-weighting applies to college basketball ratings

- **Volatility regime detection**: Our 6-factor model with novel Volatility Factor and HMM regime switches transfers to modeling tournament vs. regular-season dynamics
- **Championship prediction**: AUC ∼0.78 on 20-year NBA data using bench depth, rotation entropy, and stagger index

2 Datasets

2.1 Primary: Kaggle March ML Mania (2014–2026)

Twelve years of structured competition data comprising ∼35 CSVs per edition. Table 1 summarizes the core files.

Table 1: Core Kaggle dataset files (M = Men's, W = Women's prefix).

File	Description
`MRegularSeasonDetailedResults`	Box scores: FGM/A, 3PM/A, FTM/A, OR, DR, Ast, TO, Stl, Blk, PF
`MNCAATourneyDetailedResults`	Tournament game box scores (same schema)
`MNCAATourneySeeds`	Seedings W01–Z16 per season
`MMasseyOrdinals`	Daily rankings from 100+ systems (KenPom, Sagarin, BPI, RPI, . . .)
`MNCAATourneySlots`	Bracket structure mapping
`MGameCities`	Game locations (travel distance features)
`MTeamCoaches`	Coaching tenure and tournament experience

The competition has evolved through four eras (Table 2), with 24 total competition instances across community, Google Cloud–sponsored, and variant-scoring formats.

Table 2: Kaggle NCAA competition history.

Era	Years	Sponsor	Format
Community	2014–2017	Kaggle	Unified, Log Loss
Google Cloud	2018–2020	Google Cloud + NCAA	M/W split + Analytics track
Return	2021–2022	Kaggle	M/W split + Spread variant
Unified	2023–2026	Kaggle ($50K)	Combined M+W, Brier Score

2.2 External Rating Systems

Table 3: External data sources for feature engineering.

Source	Key Metrics	Cost	Access
KenPom	AdjO, AdjD, AdjEM, AdjT, Luck, SOS	$24.95/yr	`kenpompy`
Barttorvik	T-Rank, Barthag, adj. efficiency	Free	Web scraping
Massey Ratings	100+ system composite	Free	In Kaggle CSV
EvanMiya	BPR (Bayesian Performance Rating)	$29.99/mo	Subscription
Sports Reference	Complete historical box scores	Free	`sports-reference.com/cbb/`

2.3 BigQuery Public Dataset

Google's `bigquery-public-data.ncaa_basketball` dataset provides play-by-play data for every D1 game since 2009 (>40M plays), accessible via the free BigQuery tier.

2.4 Prediction Markets

Table 4: Prediction market platforms.

Platform	Maker Fee	API	Markets
Kalshi	0%	`docs.kalshi.com`	Championship, regional, game-by-game, props
Polymarket	Rebates	`py-clob-client`	450+ NCAA markets (non-US only)
DraftKings	—	Odds API	Spreads, totals, moneylines

Key market makers: SIG/Nellie Analytics (Kalshi's first institutional MM), DRW ("Information Finance" desk), Jump Trading (equity stake in Kalshi), Jane Street (dedicated sports quant team).

3 Methodology

3.1 Feature Engineering

We construct features in four layers of increasing sophistication:

Layer 1: Static Team Ratings. Pre-tournament snapshots of KenPom AdjEM, Barttorvik T-Rank, Massey composite rank, seed number, and conference strength (avg. conference AdjEM).

Layer 2: Matchup Features. For each pair (A, B): seed difference Δs, rating differences $\Delta r_{\text{POM}}, \Delta r_{\text{SAG}}, \Delta r_{\text{MOR}}$, tempo mismatch $|T_A - T_B|$, offensive–defensive style clash (O_A vs. D_B), geographic distance to game site, and coach tournament experience.

Layer 3: Temporal / Momentum. Rolling 10-game win percentage, average margin, late-season rating trend (Δr over last 30 days), and conference tournament momentum.

Layer 4: Market. Market-implied probability from Kalshi/Vegas, model–market disagreement signal $|p_{\text{model}} - p_{\text{market}}|$.

3.2 Model Architecture

We employ a three-tier model hierarchy (Figure **??** conceptual).

3.2.1 Tier 1: Baselines

1. **Seed Logistic**: $P(A \text{ wins}) = \sigma(\beta_0 + \beta_1 \cdot \Delta s)$ — the simplest meaningful model
2. **Elo Ratings**: Season-long Elo with tuned K-factor, home-court advantage, and inter-season regression
3. **KenPom Logistic**: Logistic regression on AdjEM difference — the standard strong baseline

3.2.2 Tier 2: Advanced Models

4. **Gradient Boosting** (XGBoost/LightGBM): Full feature set with walk-forward cross-validation. Expected to be the strongest single model based on competition history.

5. **Mixture-of-Experts (MoE)**: The core methodological contribution. A gating network $g(\mathbf{x})$ classifies each matchup into K game types, and specialized expert models $f_k(\mathbf{x})$ handle each:
$$P(A \text{ wins} \mid \mathbf{x}) = \sum_{k=1}^{K} g_k(\mathbf{x}) \cdot f_k(\mathbf{x}) \tag{2}$$
Trained via EM: alternate between soft cluster assignment and expert parameter updates. Hypothesized game types:
   - *Chalk* (1–4 vs. 13–16): High-seed dominance, rare upsets
   - *Competitive* (5–8 vs. 9–12): Upset-prone, seed provides weak signal
   - *Toss-up* (close seeds): Near coin-flip, momentum/matchup matters
   - *Cinderella* (mid-major vs. power): Conference mismatch dynamics
6. **Transformer Sequence Model**: Encode each team's season as a sequence of game embeddings:
$$\mathbf{h}_{\text{team}} = \text{TransformerEncoder}(\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_T)[\texttt{CLS}] \tag{3}$$
where $\mathbf{e}_t$ encodes game t's opponent rating, margin, location, and box score. Matchup prediction via MLP on $[\mathbf{h}_A; \mathbf{h}_B]$. Architecture: 4 layers, 64-dim, 4 heads. Pretrained on regular-season games, fine-tuned on tournament outcomes.

3.2.3 Tier 3: Meta-Models

7. **Weighted Ensemble**: $\hat{p} = \sum_m w_m \hat{p}_m$, weights $\{w_m\}$ optimized on holdout Brier score via constrained optimization ($w_m \geq 0$, $\sum w_m = 1$).
8. **Market-Calibrated Ensemble**: Bayesian update of ensemble prediction $\hat{p}$ with market-implied probability p_{mkt}:
$$p_{\text{final}} = \frac{\hat{p}^{\alpha} \cdot p_{\text{mkt}}^{1-\alpha}}{\hat{p}^{\alpha} \cdot p_{\text{mkt}}^{1-\alpha} + (1-\hat{p})^{\alpha} \cdot (1-p_{\text{mkt}})^{1-\alpha}} \tag{4}$$
where $\alpha \in [0, 1]$ controls trust in model vs. market (tuned on backtest).
9. **Portfolio Optimizer** (Kaggle-specific): Generate N bracket portfolios maximizing $P(\text{top-}K \text{ finish})$ given ensemble probabilities and estimated competitor distribution [Landgraf, 2017].

3.3 Kalshi Trading Strategy

- **Signal**: Trade when $|p_{\text{model}} - p_{\text{market}}| > \tau$ (threshold tuned on backtest)
- **Sizing**: Fractional Kelly criterion ($f = 0.25 \cdot f^*_{\text{Kelly}}$) for bankroll safety
- **Execution**: Limit orders via Kalshi REST API (0% maker fee)
- **Risk**: Max 5% bankroll per game, max 20% total exposure

4 Evaluation

4.1 Backtest Protocol

Leave-one-tournament-out (LOTO) cross-validation on 12 tournaments (2014–2026). For each holdout season s:

1. Train on all seasons $\neq s$ (regular season + tournament)
2. Predict all tournament matchups in season s
3. Compute Brier score on $\sim$63 games

Report mean $\pm$ std Brier across 12 folds.

Table 5: Target Brier score benchmarks.

Model	Brier Score
Coin flip (0.5 for all)	0.250
Seed baseline	∼0.200
KenPom logistic	∼0.185
Vegas line	∼0.175
Our target	**<0.170**

4.2 Benchmarks

4.3 Calibration

We evaluate calibration via reliability diagrams: bin predictions by decile and compare predicted probability to observed win frequency. A well-calibrated model should lie on the $y = x$ diagonal.

5 Timeline

Table 6: Project timeline (6 weeks, March 20 – May 1, 2026).

Week	Date	Deliverable
1–2	Mar 20 – Apr 3	Data pipeline, EDA notebook, feature matrix
2–3	Apr 3 – Apr 10	Baseline models (Elo, Logistic, XGBoost), LOTO backtest
3–4	Apr 10 – Apr 17	MoE implementation + Transformer sequence model
4–5	Apr 17 – Apr 24	Full 12-year backtest, ensemble optimization
5–6	Apr 24 – May 1	Kalshi API integration, paper draft, Kaggle 2027 pipeline

6 Risk Assessment

Table 7: Key risks and mitigations.

Risk	Impact	Mitigation
63 games insufficient for DL	High	Heavy regularization; pretrain on regular season (∼5,000 games/yr)
KenPom paywall changes	Medium	Barttorvik (free) as primary fallback
Kalshi regulatory shutdown	Medium	Backtest-only mode; DraftKings as alternative
Simple models dominate	Medium	Always benchmark; value is in MoE insight, not raw Brier
Single-tournament luck	High	Evaluate on 12-year LOTO, not single tournament

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